

Scientist Steve Howell in 1986 created a tobacco plant that can glow in dark like a firefly. He isolated a gene called luciferase which are present in fireflies and are the reason for the fireflies to glow in dark. He introduced this gene into toboco plants and showed that they can also glow in dark. These type of plants which contains foreign genes are known as transgenic plants or genetically modified (GM) plants. The methods used for producing such organisms are collectively known as genetic engineering. These methods promise increased agricultural yields by producing transgenic plants which can be pest resistant, drought resistant, saline resistant etc., However extreme caution has to be employed when introducing these plants to main stream agriculture as several factors has to be considered such as the balance in eco system, cost and effectiveness of the GM variety while cultivating in the field etc., Some countries have also banned GM crops and many countries follow the policies of strictly regulated use.



Har Gobind Khorana was an Indian-American biochemist who did pioneering experiments to reveal the genetic code. He was born to Krishna Devi Khorana and Ghanpat Rai Khorana in Punjab during 1922. He finished his Bachelaor's and Master's degrees in Punjab University (Lahore). He lived in pre independent India until 1945, after which he moved to England and obtained his PhD in organic chemistry from The University of Liverpool. In 1960 he was appointed as a professor in University of Wisconsin (USA) where he did a serried of experiments to understand how RNA codes for proteins. He synthesized repeating units of nucleotides 'UCU CUC UCU ...' codes for the amino acids serine and leucine repeatedly. By this way he was able to show that three RNA nucleotides were responsible for an amino acid. He also identified stop codons. In short he was able to synthesize a gene or in other words 'a nucleotide sequences which could code for a desired aminoacid sequence' He was awarded Nobel prize in 1968 as recognition for his works in understanding the molecular biology of protein synthesis.

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